

Gold-Free Faithfulness Metrics for NL-to-FOL Translation

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Abstract

Translating natural language into first-order logic (NL-FOL) is a prerequisite for neurosymbolic reasoning pipelines, yet evaluating whether a candidate formula faithfully captures a sentence’s meaning remains an open problem. Gold FOL annotations are expensive, non-unique, and unreliable: approximately 42% of entries in both FOLIO and MALLS contain incorrect formalisations. This report investigates gold-free metrics that predict faithfulness to the text given only a sentence and a candidate formula, with no reference formula, no ontology, and no domain knowledge. Over five iterations, we screen four metric families (FOL-Triage, round-trip verification, cross-system consensus, and LLM judges), construct development and confirmation datasets, and quantify a fundamental rename tradeoff. The cross-family consensus metric (`c_score_align`) achieves stratified AUROC 0.742 on a 700-sentence development set, beating the best cheap LLM judge (0.641) by +0.101 [+0.053, +0.145]. On controlled-vocabulary composed sentences, it reaches within-template AUROC 0.954 with zero error endorsement. The rename tradeoff—vocabulary dependence as the central unsolved problem—remains open: any renaming of predicates breaks the consensus signal (RENAME_NONCE false-alarm rate 0.765), and no name-free variant simultaneously achieves low rename FA and high discrimination. All confirmation arms were budget-stopped before reaching testability.

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1 Framing

Translating natural language into first-order logic (NL-FOL) is a prerequisite for neurosymbolic reasoning pipelines (Pan et al., 2023), yet evaluating whether a candidate formula faithfully captures a sentence’s meaning remains an open problem. Gold FOL annotations are expensive, non-unique, and unreliable: Brunello et al. (2026) found that approximately 42% of entries in both FOLIO (Han et al., 2022) and MALLS (Yang et al., 2023a) contain incorrect formalisations. Existing metrics either require a gold formula (exact match, prover-checked equivalence) or only verify that the output parses (Thatikonda et al., 2025, 2024). The user’s own pilot study used structural consistency checks (arity agreement, predicate-set overlap across reruns, joint satisfiability) that measure stability but not whether a formula says what its sentence says.

This run investigates gold-free metrics that predict faithfulness to the text given only a sentence and a candidate formula, with no reference formula, no ontology, and no domain knowledge. The evaluation is a meta-evaluation: we build a labelled set of (sentence, formula) pairs from public benchmarks whose gold has been audited, and measure each metric’s ability to discriminate correct translations from errors.

2 Iteration 1

2.1 Strategy

The iteration screens four candidate metric families head-to-head on a frozen evaluation set, with a disguised cheap LLM judge as the bar to beat.

Candidates. (A) FOL-Triage, a three-layer cascade: L1 (formula lint: deterministic formula smells), L2 (text accounting: bag-of-words and role-aware coverage), and L3 (role questionnaire: a small LLM reads the text and z3 reads the formula; their structured answers are compared). (B) Round-trip verification: verbalise the formula back to natural language and compare to the original via NLI or embedding similarity (Amrollahi et al., 2026). (C) Cross-system consensus: collect translations of the same sentence from multiple independent systems, measure z3-equivalence between the candidate and the peer pool, and score disagreement (Li et al., 2024). (D) Disguised cheap judge and baselines: an LLM judge, structural pilot metrics, and round-trip features evaluated on nonce-disguised sentences to control for memorisation.

Evaluation set. The frozen screen uses Logic-LM (Pan et al., 2023) released outputs on FOLIO validation: three LLM systems (GPT-3.5-turbo, GPT-4, text-davinci-003) producing 796 items. Track L contains LLM outputs labelled by z3 equivalence to corrected gold (Brunello et al., 2025); track H contains the 294 original human gold formulas, of which 57 are errors according to the corrected labels.

Decision rule. A candidate must achieve AUROC ≥ 0.65 on track L (ERROR vs CORRECT), with coverage $\geq 90\%$, rewrite false-alarm rate $\leq 10\%$ per rewrite family, and cost $\leq \$0.002$ per item.

Development dataset E (artifact `gen_art_dataset_1`). A separate 700-sentence, 8,507-row development set was constructed from FOLIO training premises, MALLS training long/conditional/exception strata, and FOLIO curated conclusions, each translated by 9 LLM families plus ccg2lambda plus GPT-4 gold. Labels were assigned by z3 equivalence to audited gold, with contested cases adjudicated by a three-member panel (Claude Haiku-4.5, GLM-4.6, Kimi-K2). The dataset contains 1,750 CORRECT, 5,005 ERROR, 1,086 UNPARSEABLE, 276 UNRESOLVED, and 390 CONTESTED items. Cost: \$9.83 in API calls.

2.2 Experiment A: FOL-Triage

[Artifact: *gen_art_experiment_1*]

FOL-Triage was evaluated on the frozen screen with Gemini 2.5 Flash Lite as the primary LLM for L3. Total API spend was \$0.29.

2.2.1 Headline AUROCs (track L, ERROR vs CORRECT, n = 477, 199 errors)

Table 1: FOL-Triage headline AUROCs on track L.

Metric	AUROC	95% CI	AUPRC
L1 (any smell)	0.508	[0.499, 0.519]	0.425
L2 bag-of-words	0.737	[0.677, 0.801]	0.690
L2 role-aware	0.557	[0.527, 0.594]	0.476
L3 score	0.756	[0.701, 0.809]	0.657
Fused (fitted on H, no leak)	0.759	[0.695, 0.825]	0.740
Fused (all H)	0.827	[0.778, 0.874]	0.771
Fused (cross-fit L)	0.864	[0.824, 0.902]	0.789
Decomposed judge	0.716	[0.645, 0.789]	0.717
L3 with local Qwen2.5-1.5B	0.708	[0.637, 0.774]	0.603

The primary no-leak fused score is 0.759. The fused-all-H and cross-fit-L numbers (0.827, 0.864) are shown for completeness but are not valid held-out estimates.

Track H (human gold errors, n = 258, 57 errors). Fused out-of-fold AUROC is 0.777 [0.698, 0.847]. L1 alone reaches 0.655, compared to 0.508 on track L.

2.2.2 L1 lint: a negative result on LLM outputs

L1 flags formulas with deterministic smells (existential implication, free variables, arity clashes, vacuity, trivial formulas, dangling predicates). On track L, the true positive rate is 2.0% at a false positive rate of 0.4%, giving AUROC 0.508. This is a negative result: L1 lint does not transfer from human annotation errors (where it is effective, AUROC 0.655) to LLM outputs.

Table 2: L1 lint false-alarm rates by smell and system (track L).

Smell	gpt-3.5 FA	gpt-4 FA	davinci-003 FA
EX_IMP	0.0	0.0	0.0
ALL_AND	0.0	0.0	0.0
IFF_RESTR	0.0	0.009	0.0
GLUE	0.0	0.0	0.0
FREE	0.0	0.0	0.0
VACUOUS	0.0	0.0	0.0
TRIVIAL	0.0	0.0	0.0
ARITY	0.0	0.0	0.0
DANGLING	0.0	0.0	0.0

2.2.3 Decomposed judge ablation

Replacing z3 with an LLM that reads the formula directly (the decomposed judge) drops L3 AUROC from 0.756 to 0.697 ($\Delta\text{AUROC} = -0.059$). The LLM gets only 57.6% of binary argument slots correct on track L and 36.4% on track H.

2.2.4 Per-type sensitivity (track L, at binary threshold)

Table 3: Per-type sensitivity on track L.

Error type	n	L1	L2-bow	L2-role	L3	Fused	Cascade
NEG	5	0.0	0.2	1.0	0.2	0.4	1.0
QUANT	2	0.0	0.0	0.5	0.0	0.5	0.5
CONN	4	0.0	0.0	0.0	0.0	0.0	0.0
DROP	142	0.028	0.479	0.127	0.197	0.599	0.606
ADD	150	0.020	0.473	0.107	0.173	0.653	0.573
SWAP	12	0.0	0.0	0.083	0.25	0.083	0.333
BIND	8	0.125	0.0	0.125	0.0	0.375	0.25
COMPOUND	187	0.118	0.428	0.166	0.160	0.599	0.578

2.2.5 Per-type sensitivity (track H)

Table 4: Per-type sensitivity on track H.

Error type	n	L1	L2-bow	L2-role	L3	Fused	Cascade
NEG	4	0.0	0.0	0.25	0.0	0.25	0.25
REV	1	0.0	0.0	0.0	0.0	0.0	0.0
QUANT	1	0.0	0.0	0.0	0.0	0.0	0.0
RESTR	1	1.0	1.0	0.0	0.0	1.0	1.0
CONN	8	0.25	0.125	0.0	0.375	0.375	0.375
MOVE	2	0.5	0.0	0.5	0.0	0.5	1.0
DROP	14	0.143	0.357	0.0	0.143	0.357	0.5
ADD	19	0.158	0.368	0.053	0.211	0.526	0.579
SWAP	2	0.0	0.5	0.0	0.5	0.5	0.5
BIND	1	1.0	1.0	0.0	0.0	1.0	1.0
UNGLUE	5	1.0	0.4	0.0	0.2	1.0	1.0
COMPOUND	33	0.303	0.182	0.242	0.182	0.545	0.636

2.2.6 Invariance under meaning-preserving rewrites

Five rewrite families were tested on 150 track-L CORRECT items.

Table 5: Rewrite false-alarm rates and flip rates (track L CORRECT).

Rewrite	n	L1 FA	L1 flip	bow FA	bow flip	role FA
RENAME	150	0.0	0.0	0.18	0.093	0.013
REORDER	34	0.0	0.0	0.029	0.0	0.059
DEMORGAN	33	0.0	0.0	0.030	0.0	0.061
CONTRA	40	0.025	0.025	0.10	0.0	0.025
PRENEX	1	0.0	0.0	0.0	0.0	0.0

Rewrite	role flip	L3 FA	L3 flip	Fused FA	Fused flip
RENAME	0.0	0.033	0.007	0.22	0.013
REORDER	0.0	0.059	0.0	0.235	0.0
DEMORGAN	0.0	0.030	0.030	0.182	0.061
CONTRA	0.025	0.025	0.0	0.275	0.0
PRENEX	0.0	0.0	0.0	0.0	0.0

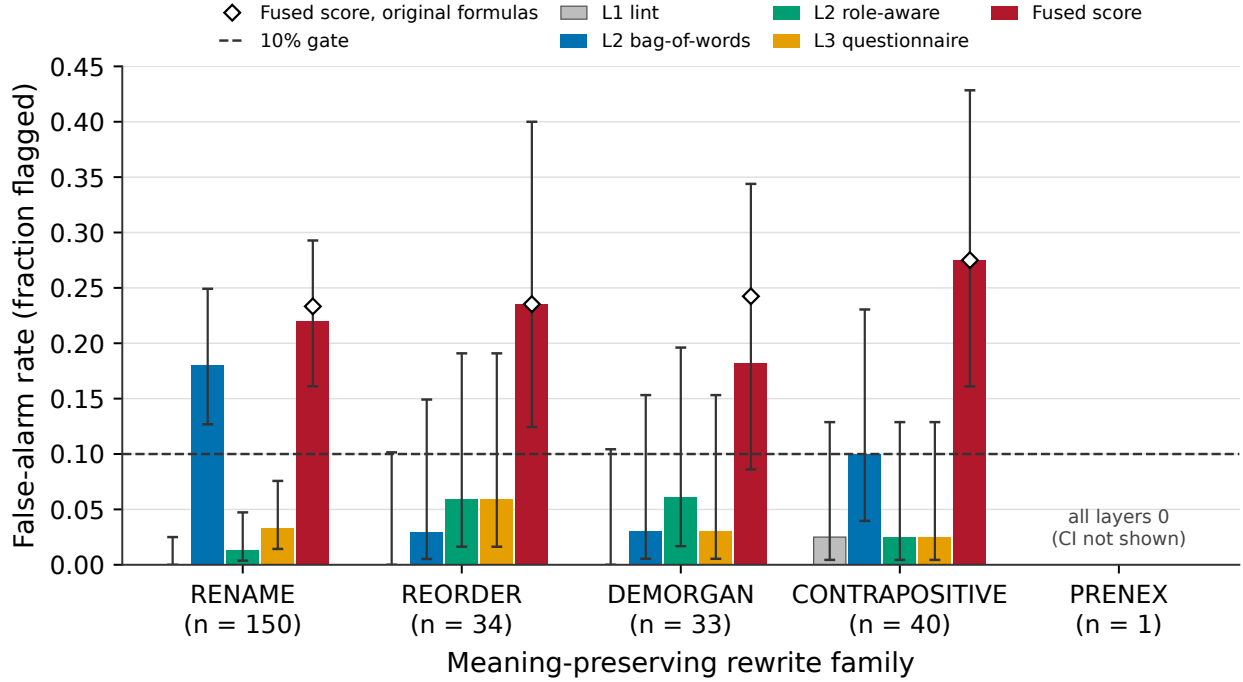


Figure 1: Rewrite false-alarm rates across metric families and rewrite types. L1, L2-role, and L3 individually pass the 10% gate. L2-bow fails only under RENAME (FA = 0.18). The fused score’s high FA is driven by calibration mismatch, not rewrite sensitivity.

Table 6: Coverage rates by track.

Track	n	Parseable	Full coverage rate
L	753	693	0.920
H	302	279	0.924

2.2.7 Coverage

2.2.8 Cost

L1 and L2 are free (CPU-only). L3 costs \$0.175 per 1,000 calls with Gemini 2.5 Flash Lite. The total per-item cost is \$0.0002, passing the \$0.002 gate. CPU time (L1 + L2 + z3): median 35 ms, P95 90 ms. L3 latency: median 630 ms, P95 1.68 s.

2.2.9 Contamination check

L3 was run on nonce-disguised sentences. AUROC with exact text-formula alignment: 0.765; with disguised text: 0.770. Paired difference: -0.005 $[-0.063, 0.049]$. There is no evidence that the LLM benefits from having seen the public benchmark sentences.

2.2.10 Complexity strata (track L, fused score)

Table 7: AUROC by complexity stratum (track L).

Stratum	n (error/correct)	Testable	Fused	L2-bow	L3
words < 12	443 (179/264)	Yes	0.781	0.742	0.772
words 12–19	31 (18/13)	No	0.521	0.596	0.541
words $\geq$ 20	3 (2/1)	No	0.500	1.000	0.000
0 quantifiers	302 (130/172)	Yes	0.743	0.728	0.771
1 quantifier	164 (65/99)	Yes	0.794	0.748	0.721
$\geq$ 2 quantifiers	11 (4/7)	No	0.750	0.768	0.607
depth $\leq$ 1	296 (124/172)	Yes	0.762	0.745	0.779
depth 2–3	173 (73/100)	Yes	0.756	0.717	0.710
depth $\geq$ 4	8 (2/6)	No	0.667	0.750	0.500
0 conditions	344 (145/199)	Yes	0.775	0.759	0.791
1–2 conditions	131 (54/77)	Yes	0.716	0.677	0.628

2.2.11 Per-system AUROC (descriptive, n = 3 systems)

Table 8: Per-system AUROCs on track L.

System	n	Error rate	Fused	L1	L2-bow	L3
gpt-3.5-turbo	143	0.378	0.689	0.500	0.650	0.722
gpt-4	177	0.367	0.700	0.496	0.716	0.720
text-davinci-003	157	0.510	0.857	0.525	0.811	0.817

2.3 Experiment C: Cross-System Consensus

[Artifact: *gen_art_experiment_3*]

The consensus metric collects translations of the same sentence from multiple independent LLM families, checks pairwise z3 equivalence, and scores each candidate by its disagreement with the pool. Total API spend was \$1.52.

2.3.1 Headline AUROCs (track L, ERROR vs CORRECT, n = 546, 249 errors)

Table 9: Consensus metric AUROCs on track L.

Metric	AUROC	95% CI	AUPRC
c_score (all peers)	0.866	[0.821, 0.906]	0.827
c_score (6 fresh peers)	0.872	[0.830, 0.911]	0.825
c_score (non-OpenAI)	0.863	[0.820, 0.903]	0.808
c_score (2 Logic-LM peers)	0.753	[0.701, 0.806]	0.659
medoid depth	0.722	[0.673, 0.774]	0.622
cluster entropy	0.773	[0.724, 0.820]	0.691
SC-5 cheap (gpt-4.1-nano)	0.725	[0.669, 0.781]	0.627
SC-5 same model	0.716	[0.632, 0.796]	0.628
SC-5 entropy	0.631	[0.568, 0.693]	0.539
Prediction set instab. (SC)	0.688	[0.626, 0.750]	0.602
Prediction set instab. (pool)	0.770	[0.717, 0.821]	0.721
FOL length	0.527	[0.470, 0.586]	0.485
Misalignment (vocab)	0.835	[0.794, 0.876]	0.821

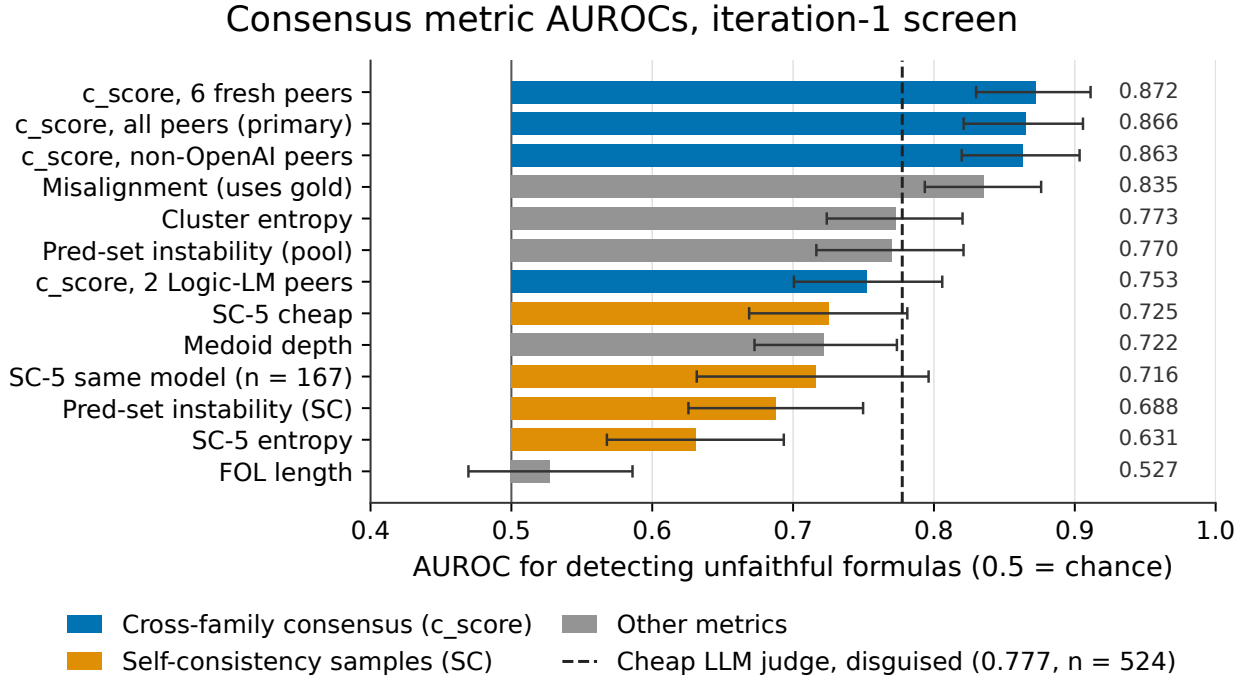


Figure 2: Consensus metric AUROCs on the frozen screen (track L). The c_score achieves the highest AUROC at 0.866, beating SC-5 by $\Delta\text{AUROC} = +0.140$ [0.085, 0.197].

The c_score achieves the highest AUROC in the iteration at 0.866. It beats SC-5 (the self-consistency baseline) by $\Delta\text{AUROC} = +0.140$ [0.085, 0.197]. Using only the six fresh peer translations (not the three Logic-LM systems) gives 0.872. Restricting to the two other Logic-LM systems drops the AUROC to 0.753, showing that diversity across model families is essential.

2.3.2 RENAME rewrite invariance: failure

Under predicate renaming, the `c_score` false-alarm rate is 96% (100% for constant-only renaming). This is expected and fundamental: two formulas with different predicate names are not z3-equivalent, so any rename breaks the equivalence check.

2.3.3 Blind-spot analysis

38% of ERROR items are z3-equivalent to the medoid (the most popular peer formula). This places a ceiling on recall of approximately 62%. The blind-spot rate is 5% for polarity errors and 57% for structural errors.

2.3.4 Cost

The peer pool costs \$0.0015 per sentence (6 peers via API). Total experiment spend: \$1.52.

2.4 Experiment D: Disguised Judge and Baselines

[*Artifact: gen_art_experiment_4*]

This experiment establishes the bar: every baseline metric and every judge variant on the same shared screen, scored with nonce-disguised sentences. Total API spend was \$2.02.

2.4.1 The bar: disguised cheap judge

The primary cheap judge (Gemini 2.5 Flash Lite, JSON 0-100 scoring, nonce-disguised sentences) achieves AUROC 0.777 [0.720, 0.827] on track L (n = 524, 230 errors, 294 correct).

2.4.2 Full baseline table (track L, ERROR vs CORRECT, n = 524)

2.4.3 Cross-fitted baseline combinations (track L)

2.4.4 Contamination analysis

No evidence of contamination was found for any judge. All DiD CIs include zero:

2.4.5 Invariance of judges under meaning-preserving rewrites

FOL-Triage has the lowest flip rate of any metric tested. Judges and round-trip methods change their verdict on 29–49% of meaning-preserving rewrites.

2.4.6 Per-error-type sensitivity of judges (track H, curator view, n = 46 errors)

[Correction (C2): on development set E at matched FA 0.10, the original judge has higher recall than the disguised judge. Disguise does not uniformly improve the judge; the screen-level pattern does not generalise.]

2.4.7 Complexity analysis (track L)

A logistic regression of judge correctness on standardised word count yields a coefficient of -0.48 (SE 0.11, $p < 0.001$): the judge is significantly less reliable on longer sentences.

Table 10: Full baseline AUROCs on track L.

Metric	AUROC	95% CI
parse_fail	0.500	[0.500, 0.500]
pilot_joint_conflict	0.513	[0.488, 0.539]
pilot_arity_incons	0.500	[0.500, 0.500]
pilot_shape_incons	0.524	[0.470, 0.578]
pilot_dangling	0.533	[0.474, 0.592]
pilot_undeclared	0.525	[0.510, 0.542]
pilot_rerun_jacc	0.720	[0.659, 0.780]
rt_nli_min	0.710	[0.648, 0.769]
rt_nli_fwd	0.652	[0.588, 0.713]
rt_nli_bwd	0.681	[0.621, 0.742]
rt_nli_contra	0.628	[0.566, 0.686]
rt_nli_min_alt	0.689	[0.627, 0.749]
rt_embed_cos	0.633	[0.574, 0.696]
rt_reformalise_eq	0.513	[0.479, 0.547]
judge_cheap_orig	0.749	[0.695, 0.801]
judge_cheap_disg	0.777	[0.720, 0.827]
judge_cheap2_orig (gpt-4.1-nano)	0.750	[0.691, 0.810]
judge_cheap2_disg	0.714	[0.648, 0.777]
judge_strong_orig (gemini-3.1-pro)	0.802	[0.715, 0.877]
judge_strong_disg	0.730	[0.655, 0.803]
judge_local_qwen8b_disg	0.757	[0.702, 0.811]
judge_local_llama8b_disg	0.671	[0.604, 0.737]
judge_local_qwen14b_disg	0.760	[0.681, 0.834]
rt_nli_min_localverb	0.740	[0.680, 0.796]

Table 11: Cross-fitted baseline combinations on track L.

Combination	AUROC	95% CI
S1 (structural)	0.698	[0.634, 0.761]
S2 (round-trip)	0.687	[0.625, 0.749]
S3 (judges disguised)	0.771	[0.705, 0.831]
S4 (all cheap features)	0.817	[0.762, 0.867]
S6 (S4 + local judges)	0.810	[0.753, 0.861]

Table 12: Contamination difference-in-differences (DiD).

Judge	DiD (H – L)	95% CI
flash-lite	−0.042	[−0.170, 0.094]
gpt-4.1-nano	−0.077	[−0.188, 0.023]
gemini-3.1-pro	+0.002	[−0.115, 0.104]
Qwen3-8B	−0.045	[−0.161, 0.079]
Llama-3.1-8B	+0.023	[−0.112, 0.157]
Qwen3-14B	−0.073	[−0.195, 0.056]

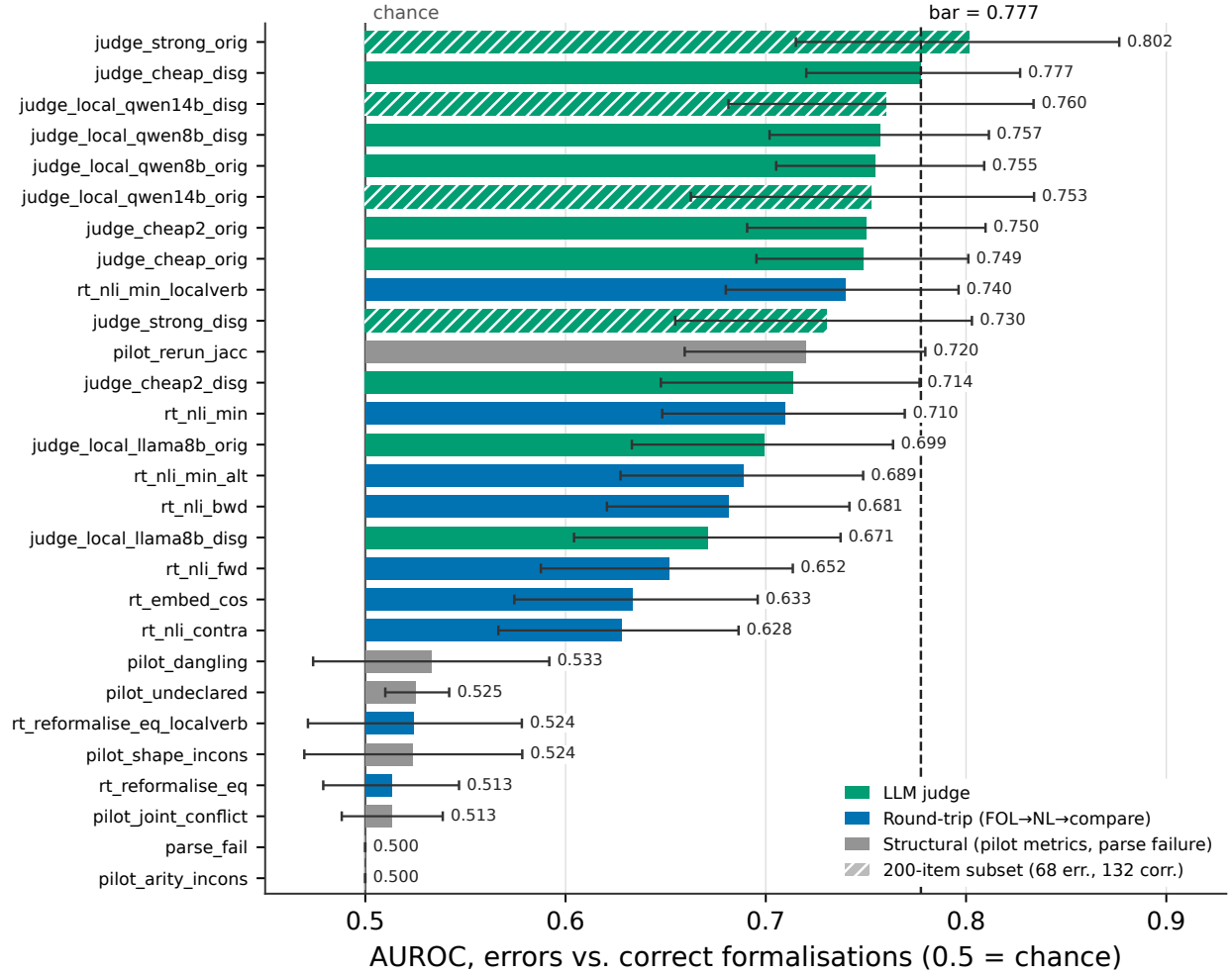


Figure 3: Baseline AUROCs on the screen (track L). The disguised cheap judge (flash-lite) sets the bar at 0.777. Pilot structural metrics are null (0.500–0.533).

Table 13: Judge flip rates under meaning-preserving rewrites (track L CORRECT, n = 270).

Metric	RENAME	REORDER	DEMORGAN	CONTRA	ALL
judge_cheap_orig	0.27	0.12	0.23	0.64	0.29
judge_cheap_disg	0.44	0.12	0.46	0.41	0.39
judge_cheap2_orig	0.49	0.12	0.46	0.95	0.49
rt_nli_min	0.37	0.10	0.31	0.33	0.31
rt_embed_cos	0.57	0.07	0.44	0.38	0.44
FOL-Triage fused	0.013	0.0	0.061	0.0	0.013

Table 14: Judge per-error-type sensitivity (track H).

Group	n	cheap_orig	cheap_disg	cheap2_orig	rt_nli_min
1-op errors	13	0.308	0.538	0.462	0.231
2-op errors	11	0.545	0.818	0.727	0.273
ADD	12	0.500	0.750	0.667	0.250
COMPOUND	22	0.545	0.864	0.773	0.273
DROP	11	0.455	0.636	0.455	0.182
CONN	3	0.667	0.667	1.000	0.333
NEG	2	0.500	0.500	1.000	0.500
SWAP	2	0.000	1.000	0.500	0.000

Table 15: Judge AUROC by complexity stratum (track L).

Stratum	n (err/cor)	judge_cheap_disg	rt_nli_min
words < 12	485 (210/275)	0.784	0.713
words 12–19	36 (18/18)	0.667	0.670
quantifiers 0–1	512 (226/286)	0.784	0.712
depth ≤ 3	466 (199/267)	0.781	0.702
depth 4–5	54 (30/24)	0.762	0.828
0 conditions	394 (182/212)	0.783	0.745
1–2 conditions	128 (48/80)	0.765	0.653

Table 16: Iteration 1 summary: four candidates and the bar.

Method	AUROC	95% CI	n	Cost/item	Rewrite flip
c_score (consensus)	0.866	[0.821, 0.906]	546	\$0.0015/sent	96% (RENAME)
S4 baseline combo	0.817	[0.762, 0.867]	524	varies	varies
Bar: judge_cheap_disg	0.777	[0.720, 0.827]	524	\$0.00004	0.39
FOL-Triage fused	0.759	[0.695, 0.825]	477	\$0.0002	0.013
rt_nli_min	0.710	[0.648, 0.769]	524	\$0.00001	0.31
L1 lint	0.508	[0.499, 0.519]	477	\$0	0.0
Pilot structural	0.50–0.53	n/a	524	\$0	n/a

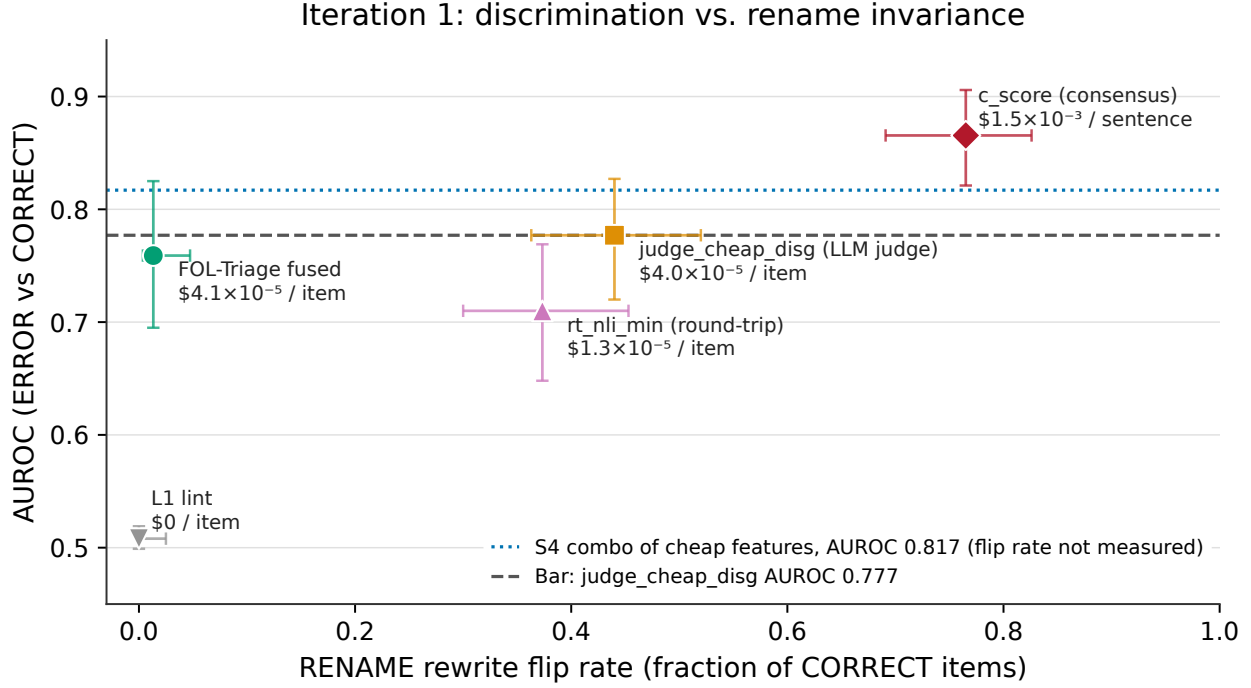


Figure 4: Iteration 1 summary comparison. Consensus leads at AUROC 0.866, followed by the S4 baseline combination (0.817) and the disguised cheap judge bar (0.777). FOL-Triage (0.759) is below the bar but has the lowest rewrite flip rate (0.013).

2.5 Iteration 1 Summary

Key findings:

1. The cross-system consensus metric ($c_score = 1 - eq_frac$) achieves the highest AUROC at 0.866, beating SC-5 by +0.140. It requires peer translations from multiple model families and is fundamentally non-invariant to predicate renaming (96% FA under RENAME).
2. The disguised cheap judge sets the bar at AUROC 0.777. The S4 combination reaches 0.817 (+0.039, not significant). The frontier judge adds nothing significant over the cheap judge under solver labels; under panel labels the advantage strengthens (+0.166, $p < 0.001$).
3. FOL-Triage achieves AUROC 0.759, below the bar. Its primary value is invariance: the fused flip rate under meaning-preserving rewrites is 0.013, compared to 0.29–0.49 for judges.
4. L1 lint does not transfer from human annotation errors to LLM outputs. The pilot structural metrics are null. Round-trip NLI is 0.067 below the judge.
5. The evaluation labels are noisy: 58.8% of track-L “errors” are pure ADD+DROP (likely vocabulary mismatches).
6. No contamination was detected for any LLM-based metric.

3 Iteration 2

3.1 Strategy

Iteration 2 has three goals: (i) confirm iteration 1’s ordering on the development dataset E under both the solver labels (R_AB, R_A) and panel-adjudicated labels (R_ADJ); (ii) test whether fusing the consensus signal (PEER) with text-based checks (TEXT = L2-bow + L3) improves over each alone; and (iii) construct a typed perturbation suite (PERTURB) for controlled sensitivity testing.

The pre-registered bar is the flash-lite judge on disguised sentences, with the local Qwen3-8B judge as fallback. The shared OpenRouter key hit its daily limit before the development-set experiment completed, making the flash-lite bar UNTESTABLE on E.

3.2 Experiment 5: PEER+TEXT on development dataset E

[Artifact: *gen_art_experiment_5*]

PEER+TEXT is a logistic fusion of the consensus signal (c_score_align) and the text-based FOL-Triage layers (L2-bow, L3 questionnaire). API spend was \$0.

3.2.1 Headline AUROCs on E (R_AB, ERROR vs CORRECT)

Table 17: PEER+TEXT headline AUROCs on development set E.

Subset	n (err/cor)	PEER+TEXT	c_align	NF c	L2-bow	L3	Local judge
R_AB pooled	2686 (1822/864)	0.790	0.782	0.743	0.734	0.579	0.712
R_AB L25	873 (697/176)	0.755	0.712	0.651	0.722	0.560	0.664
R_AB L20	821 (592/229)	0.747	0.722	0.643	0.715	0.547	0.683
R_AB EXC	606 (384/222)	0.777	0.814	0.782	0.669	0.620	0.663
R_AB CTRL	386 (149/237)	0.708	0.742	0.741	0.613	0.490	0.739
R_AB long pool	2300 (1673/627)	0.768	0.759	0.704	0.717	0.580	0.653
R_A L20+EXC	449 (261/188)	0.818	0.860	0.802	0.678	0.587	0.554

The paired Δ AUROC of PEER+TEXT minus the local judge is +0.078 [0.043, 0.112] on R_AB pooled (DeLong $p < 0.0001$). On the long pool, the gap widens to +0.115 [0.075, 0.157].

The stratified AUROC (pairing only within complexity strata) shows PEER+TEXT at 0.753 and c_score_align at 0.741, a difference of +0.011 [−0.012, 0.038], not significant. The TEXT layers add no significant discriminative power over c_score_align alone once the stratum-composition confound is removed.

3.2.2 Mechanism predictions

P1 (PEER catches COVERAGE errors that TEXT misses, and vice versa). At matched false-alarm rate 0.10: PEER has higher recall than TEXT on COVERAGE errors (+0.093 [0.046, 0.147]), and TEXT has higher recall on PEER_ENDORSED errors (+0.103 [0.051, 0.171]). Verdict: INCONCLUSIVE.

P2 (PEER and TEXT error rankings are uncorrelated). Spearman correlation between PEER and TEXT among errors is 0.294 [0.224, 0.365]. Verdict: REFUTED.

P3 (PEER+TEXT advantage grows with sentence complexity). The logistic interaction term is −0.023 [−0.228, 0.185], consistent with no interaction. Verdict: INCONCLUSIVE.

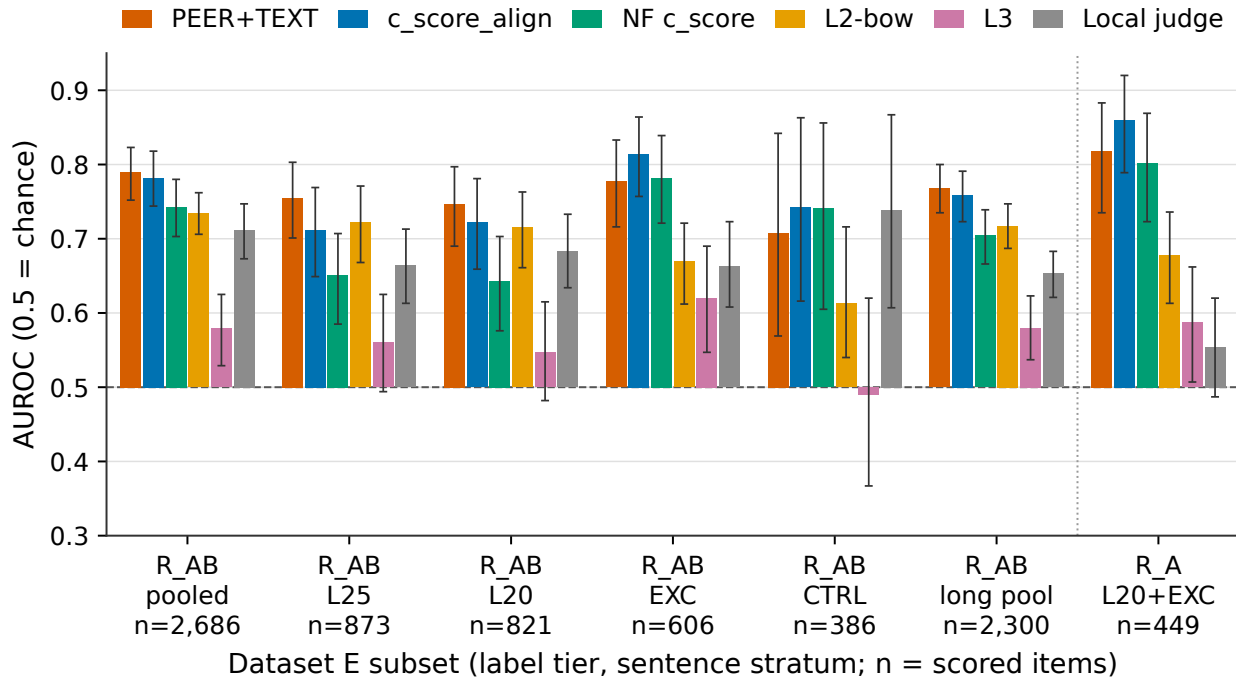


Figure 5: AUROCs on development set E by metric and subset. PEER+TEXT leads on R_AB pooled (0.790); the local judge lags at 0.712.

P4 (PEER+TEXT is more invariant to meaning-preserving rewrites than judges).
 RENAME FA: fused 0.385, local judge 0.683. Verdict: REFUTED (c_score_align drives most of the FA at 0.859).

3.2.3 Rewrite false alarms (screen, CORRECT bases)

Table 18: Rewrite false-alarm rates on screen CORRECT bases.

Rewrite	n	Fused	c.align	NF g	L2-bow	L3	Local judge
RENAME	104	0.385	0.859	0.074	0.216	0.168	0.683
REORDER	27	0.222	0.178	0.037	0.044	0.444	0.000
DEMORGAN	26	0.154	0.185	0.038	0.046	0.231	0.231
CONTRA	28	0.107	0.063	0.237	0.075	0.188	0.214

3.3 Evaluation 1: Re-evaluation under alternative label regimes

[Artifact: *gen_art_evaluation_1*]

All 57 reviewer-audited quantities from the iteration-1 report reproduce exactly under independent recomputation (57/57 MATCH).

Table 19: Regime AUROC matrix on common item set.

Metric	R_SOLVER	R_ADJ_AB	R_ADJ_A	R_ADJ_ALL
PT (PEER+TEXT)	0.871	0.877	0.932	0.906
fused_H (FOL-Triage)	0.770	0.868	0.958	0.866
c_score	0.849	0.779	0.786	0.831
L2-bow	0.732	0.815	0.917	0.810
L3	0.761	0.718	0.813	0.751
judge_cheap_disg	0.782	0.772	0.751	0.816
judge_cheap_orig	0.754	0.797	0.858	0.785
sc5_cheap	0.713	0.703	0.676	0.737
rt_nli_min	0.709	0.749	0.731	0.760

3.3.1 Regime AUROC matrix (common set, selected metrics)

Under R_ADJ_AB, PEER+TEXT (0.877) leads, followed by FOL-Triage fused (0.868) and L2-bow (0.815). The c_score drops from 0.849 (solver) to 0.779 (R_ADJ_AB), because adjudicated labels reclassify vocabulary-mismatch items as CORRECT.

3.3.2 Label regime shifts

Table 20: Label regime shift: solver to adjudicated.

Metric	Shift	95% CI	Direction
fused_H	+0.080	[0.003, 0.171]	UP
L2-bow	+0.102	[0.018, 0.192]	UP
c_score	-0.106	[-0.255, 0.013]	INSIDE CI
judge_cheap_disg	+0.018	[-0.094, 0.134]	INSIDE CI

3.4 Dataset 3: R_COMP and PERTURB

[Artifact: *gen_art_dataset_3*]

3.4.1 PERTURB: complete

3.5 Experiment 6: Baselines on development set E

[Artifact: *gen_art_experiment_6*, iteration 2]

3.6 Dead ends: B2 and R_ADJ

Candidate B2 (decomposed z3 distinguishing-worlds reader): DROPPED. Gate balanced accuracy 0.476 (target ≥ 0.85). The b2_score AUROC on E (R_AB pooled) was 0.574.

R_ADJ adjudicator gate: DROPPED.

Table 21: PERTURB suite statistics.

Statistic	Value
Base formulas	300
Total mutants (DOWN + UP)	4,234
Matched controls	868
Total suite rows	5,102
Complete DOWN/UP pairs	1,354
z3 verification failures	0
Operator types	12
Cost	\$0.41

Table 22: Pooled AUROCs on E (R_AB, selected metrics).

Metric	AUROC [95% CI]
S4_local (all tiers)	0.754 [0.715, 0.789]
S4_local (R_AB)	0.748 [0.709, 0.782]
judge_local_qwen8b_disg	0.710 [0.674, 0.746]
S4_local_noLLMjudge	0.703 [0.660, 0.742]
sc5_local_eq_frac	0.664 [0.617, 0.706]
pilot_rerun_jacc	0.660 [0.621, 0.695]
judge_local_qwen8b_orig	0.649 [0.604, 0.693]
judge_local_llama8b_disg	0.636 [0.590, 0.681]
rt_nli_min_local	0.626 [0.580, 0.673]
b2_score	0.574 [0.529, 0.619]
pilot_shape_incons	0.504 [0.499, 0.510]
parse_fail	0.500 [0.500, 0.500]

Table 23: R_ADJ adjudicator gate results.

Adjudicator	Gate track-H balanced accuracy	Target
Sonnet-5	0.662	≥ 0.80
Grok-4.20	0.527	≥ 0.80

3.7 Iteration 2 Summary

1. PEER+TEXT achieves AUROC 0.790 [0.75, 0.82] on development dataset E (R_AB pooled, $n = 2686$), beating the local judge by +0.078 [0.043, 0.112] (DeLong $p < 0.0001$).
2. Fusion does not significantly beat c_score_align alone. The stratified AUROC difference is +0.011 [−0.012, 0.038].
3. The name-free consensus variant (NF-anchored) failed the pre-registered screen gate due to a threshold-tie artefact.
4. Under adjudicated labels, FOL-Triage fused rises from 0.770 to 0.868. The c_score drops from 0.849 to 0.779.
5. The frontier judge advantage is large under adjudicated labels (+0.166 AUROC, $p < 0.001$).
6. All 57 reviewer-audited numbers from iteration 1 reproduce exactly (57/57 MATCH).
7. The PERTURB suite (4,234 typed mutants + 868 controls, 12 operator types) is complete.

4 Iteration 3

4.1 Strategy

Iteration 3 resolves two gaps: the flash-lite API bar and the R_COMP candidate generation. It runs five pre-registered tests (T1–T5).

4.2 Test T1: Head-on API bar on development set E

[Artifact: *gen_art_experiment_6*, iteration 3]

4.2.1 Verdict

Table 24: T1 verdict summary.

Criterion	Status	Key numbers
(a) c_score_align beats flash-lite disg	CONFIRMED	R_AB strat Δ +0.099 [+0.049, +0.146]
(b) adds signal beyond S4_full	CONFIRMED	strat Δ +0.039 [+0.021, +0.057]; perm. null p95 = 0.009
(d) frontier ratio ≥ 0.95 at $\leq 10\times$ cost	CONFIRMED	ratio 0.957 [0.887, 1.034]; 30.2 $\times$ cheaper
VEX confound control	CONFIRMED	strat Δ +0.306 [+0.232, +0.376]
M3 length slope	DISCONFIRMED	words Δ slope +0.146 [−0.051, +0.361]

Table 25: Item-level AUROCs on E (R_AB pooled).

Metric	Pooled [CI]	Strat [CI]	AUPRC	\$/item
p_peer_text	0.790 [0.752, 0.825]	0.753 [0.720, 0.784]	0.879	1.21e-04
c_score_align	0.782 [0.745, 0.816]	0.741 [0.705, 0.775]	0.861	1.21e-04
S4_full_loof	0.777 [0.739, 0.809]	0.735 [0.699, 0.766]	0.874	–
S4_local_loof	0.748 [0.709, 0.782]	0.693 [0.656, 0.729]	0.856	–
nf_c_score	0.743 [0.702, 0.780]	0.686 [0.647, 0.720]	0.829	–
L2-bow	0.734 [0.705, 0.762]	0.697 [0.667, 0.726]	0.830	–
judge_local_disg	0.710 [0.674, 0.746]	0.674 [0.646, 0.703]	0.815	–
judge_cheap2_orig	0.700 [0.659, 0.739]	0.653 [0.616, 0.688]	0.823	2.01e-05
judge_cheap_disg	0.696 [0.655, 0.733]	0.642 [0.607, 0.676]	0.786	4.88e-05
judge_cheap2_disg	0.681 [0.641, 0.719]	0.644 [0.609, 0.680]	0.806	2.01e-05
sc5_eq_frac	0.656 [0.609, 0.699]	0.598 [0.559, 0.637]	0.758	2.95e-05
rt_nli_min	0.639 [0.593, 0.687]	0.606 [0.569, 0.645]	0.776	2.75e-05
judge_cheap_orig	0.635 [0.595, 0.671]	0.623 [0.593, 0.651]	0.772	4.88e-05
rt_nli_fwd	0.674 [0.631, 0.716]	0.611 [0.573, 0.650]	0.812	–
L3	0.579 [0.532, 0.625]	0.562 [0.524, 0.601]	0.720	–
rt_embed_cos	0.574 [0.527, 0.623]	0.566 [0.526, 0.607]	0.731	–
parse_fail	0.500 [0.500, 0.500]	0.500 [0.500, 0.500]	0.678	–

Table 26: Paired deltas: challenger minus flash-lite disguised (stratified).

Cell	n (E/C)	Challenger	Δ vs bar [CI]
R_AB pooled	1822/864	c_score_align	+0.099 [+0.049, +0.146]
R_AB pooled	1822/864	p_peer_text	+0.110 [+0.065, +0.154]
R_AB long	1673/627	c_score_align	+0.116 [+0.070, +0.163]
R_AB long	1673/627	p_peer_text	+0.132 [+0.091, +0.173]
R_AB EXC	384/222	c_score_align	+0.197 [+0.124, +0.269]
R_AB EXC	384/222	p_peer_text	+0.160 [+0.091, +0.232]
R_A L20+EXC	261/188	c_score_align	+0.299 [+0.212, +0.390]
R_A L20+EXC	261/188	p_peer_text	+0.264 [+0.180, +0.351]
R_AB CTRL	149/237	c_score_align	−0.069 [−0.232, +0.098]

Table 27: Cross-fitted stacks on E.

Stack	Pooled OOF [CI]	Strat OOF [CI]
S4_local	0.748 [0.709, 0.782]	0.693 [0.656, 0.729]
S4_full	0.777 [0.739, 0.809]	0.735 [0.699, 0.766]
S4_full + c_score_align	0.807 [0.772, 0.838]	0.774 [0.743, 0.802]
S4_full + p_peer_text	0.808 [0.775, 0.839]	0.776 [0.745, 0.805]
S4_full + g_c (all consensus)	0.809 [0.774, 0.840]	0.776 [0.745, 0.805]
S4_local + c_score_align	0.796 [0.759, 0.828]	0.757 [0.723, 0.788]
PT_refit (4 features)	0.808 [0.769, 0.842]	0.778 [0.742, 0.813]

Table 28: Nested contrasts on E.

Nested contrast	Strat Δ [CI]	Pooled Δ [CI]
S4_full + c_align - S4_full	+0.039 [+0.021, +0.057]	+0.030 [+0.015, +0.045]
S4_full + p_peer_text - S4_full	+0.041 [+0.022, +0.060]	+0.032 [+0.017, +0.046]
S4_full + nf_c_score - S4_full	+0.013 [+0.002, +0.023]	+0.010 [+0.001, +0.019]
S4_local + c_align - S4_local	+0.064 [+0.042, +0.086]	+0.048 [+0.030, +0.067]
S4_full - S4_local	+0.042 [+0.022, +0.063]	+0.029 [+0.013, +0.045]

Table 29: VEX confound control.

Subset	Strat Δ c_align - judge_cheap_disg [CI]
VEX	+0.306 [+0.232, +0.376]
VEX_STRICT	+0.202 [+0.144, +0.269] (pooled)
VEX_tierA_only	+0.224 [+0.161, +0.297] (pooled)

4.2.2 Item-level AUROC on R_AB (E_POOL PRIMARY)

4.2.3 Paired deltas: challenger minus bar (stratified)

4.2.4 Cross-fitted stacks and nesting

4.2.5 VEX confound control

4.2.6 AUROC by complexity bin

Table 30: AUROC by word-count bin.

Bin	n (E/C)	c_align	p_peer_text	judge_disg	S4_full	Δ c - judge [CI]
<12	154/283	0.725	0.746	0.766	0.772	-0.040 [-0.178, +0.103]
12-19	255/134	0.701	0.682	0.607	0.667	+0.094 [+0.013, +0.188]
20-24	686/262	0.727	0.740	0.625	0.715	+0.103 [+0.026, +0.171]
25-34	614/161	0.705	0.753	0.618	0.727	+0.087 [-0.006, +0.178]
≥ 35	113/24	0.759	0.756	0.706	0.789	+0.054 [-0.115, +0.173]

Table 31: AUROC by exception type.

Bin	n (E/C)	c_align	judge_disg	Δ [CI]
None	1393/617	0.776	0.731	+0.046 [+0.007, +0.089]
except	62/71	0.964	0.600	+0.365 [+0.236, +0.472]
unless	179/98	0.801	0.713	+0.088 [+0.001, +0.164]

The consensus advantage is concentrated in sentences with 12+ words and is largest on exception-clause sentences (AUROC 0.964 vs 0.600 for “except” sentences).

Table 32: Recall per error group at matched FA 0.10.

Group	n	c_align	p_peer_text	judge_disg	judge_strong	S4_full
COMPOUND (tier B)	1148	0.55	0.63	0.25	0.67	0.56
polarity	434	0.53	0.56	0.26	0.74	0.53
structural	1282	0.49	0.53	0.24	0.60	0.50
COVERAGE	246	0.27	0.26	0.13	0.27	0.20
MEANING_RENAME	221	0.07	0.10	0.17	0.09	0.17
PEER_ENDORSED	398	0.00	0.06	0.16	0.07	0.17
PEER_ENDORSED_NF	340	0.05	0.11	0.13	0.15	0.21

4.2.7 Recall per error group at matched FA 0.10

4.2.8 System-level correlation (10 families)

Table 33: System-level Kendall τ -b.

Metric	τ -b over sys $\times$ variant [CI]	τ -b over families [CI]
p_peer_text	0.718 [0.590, 0.846]	0.822 [0.644, 0.911]
c_score_align	0.692 [0.513, 0.795]	0.822 [0.600, 0.911]
judge_cheap2_disg	0.718 [0.590, 0.846]	0.822 [0.644, 0.911]
judge_cheap_disg	0.564 [0.410, 0.718]	0.600 [0.422, 0.778]
S4_full_oof	0.590 [0.538, 0.769]	0.644 [0.467, 0.778]

4.2.9 Mechanism predictions (T4)

[Artifact: *gen_art_evaluation_2*]

48 numerical claims from the iteration-2 report were independently recomputed. All 48 match (48/48 VERIFIED_MATCH).

M1 (peer-endorsement rate among errors falls with n_conditions): **INCONCLUSIVE**.

M2 (divergence among CORRECT rows rises with words): **CONFIRMED but non-specific** (shuffled-label placebo also passes).

M3 (consensus length slope less negative than the judge’s): **INCONCLUSIVE** (two specifications conflict).

M4 (k needed to reach 95% of full-pool AUROC is ≤ 5 families): **CONFIRMED**, k95 = 3 overall; by words tercile k95 = 2 / 3 / 5.

Table 34: k-curve on development set E.

k	AUROC	AUROC words T3	\$/sentence (FULL)
1	0.685	0.581	\$0.000315
3	0.757	0.651	\$0.000946
5	0.776	0.698	\$0.001576
7	0.784	0.726	\$0.002206

4.3 Test T2: R_COMP (long composed sentences)

[Artifact: *gen_art_experiment_7*]

221 sentences produced 2,024 candidate translations. Population: 429 ERROR / 1,595 CORRECT.

4.3.1 SIG: primary test

Table 35: SIG primary test (criterion c).

Statistic	c_score_sig	judge_cheap_disg	Δ [95% CI]	PASS
Within-template AUROC	0.954	0.587	+0.367 [+0.328, +0.404]	True

The consensus metric’s within-template AUROC is 0.954, massively outperforming the flash-lite disguised judge at 0.587. The delta of +0.367 is the largest effect observed in any test.

4.3.2 SIG: per-metric AUROCs

Table 36: SIG per-metric AUROCs on R_COMP.

Metric	n	Within-template [CI]	Pooled [CI]	AUPRC
g_align	2024	0.977 [0.970, 0.983]	0.978 [0.972, 0.983]	0.878
g_nf	2024	0.976 [0.968, 0.983]	0.976 [0.970, 0.982]	0.877
c_score_sig	2024	0.952 [0.937, 0.965]	0.964 [0.957, 0.971]	0.803
c_score_align	2024	0.952 [0.937, 0.965]	0.964 [0.957, 0.971]	0.803
judge_frontier_orig	60	0.932 [0.782, 1.000]	0.932 [0.855, 0.990]	0.881
judge_frontier_disg	60	0.729 [0.573, 0.867]	0.735 [0.593, 0.851]	0.655
judge_cheap_orig	1932	0.673 [0.637, 0.708]	0.710 [0.678, 0.741]	0.345
judge_cheap_disg	1904	0.587 [0.551, 0.622]	0.578 [0.537, 0.617]	0.238
judge_local_disg	2024	0.498 [0.459, 0.534]	0.541 [0.505, 0.575]	0.233

4.3.3 Rename invariance on R_COMP

Table 37: Rename invariance on R_COMP SIG.

Metric	Unrenamed FA	RENAME_NONCE FA	RENAME_SYN FA
c_score_sig	0.260	1.000	1.000
c_score_align	0.260	1.000	0.635
c_score_nf	0.263	0.261	0.252
c_score_hyb	0.260	0.261	0.252

4.3.4 FREE condition: NOT_TESTABLE

The FREE condition produced only 34 CORRECT tier-A items against 420 ERROR. FREE is NOT_TESTABLE and deferred to iteration 4.

4.4 Tests T3 + T5: PERTURB sensitivity and rename tradeoff

[Artifact: *gen_art_experiment_8*]

4.4.1 T3: Per-operator within-base AUROC (E_bases)

Table 38: Per-operator within-base AUROC on PERTURB.

Operator	n	c_align	c_hyb	c_nf	rt_nli_min	judge_disg
NEG	379	0.869	0.884	0.843	0.873	0.723
REV	179	0.860	0.869	0.826	0.578	0.697
QUANT	179	0.860	0.872	0.829	0.351	0.632
RESTR	177	0.859	0.873	0.827	0.608	0.638
CONN	327	0.861	0.870	0.832	0.563	0.663
DROP	191	0.841	0.841	0.791	0.792	0.728
ADD	472	0.859	0.860	0.820	0.895	0.683
SWAP	116	0.882	0.910	0.863	0.454	0.529
BIND	314	0.865	0.858	0.798	0.833	0.672
M_RENAME	318	0.872	0.685	0.499	0.930	0.600

4.4.2 T5: Rename tradeoff table (E_bases)

Table 39: Rename tradeoff: false-alarm rates and recall.

Metric	NONCE FA	SYN FA	M.RENAME wb	SWAP recall	BASE FA
c_align	0.765	0.565	0.872	0.758	0.201
c_nf	0.132	0.201	0.499	0.512	0.163
c_hyb	0.216	0.298	0.685	0.832	0.197
g_align	0.413	0.318	0.841	0.426	0.154
g_nf	0.132	0.176	0.477	0.428	0.154
rt_nli_min	0.686	0.191	0.930	0.153	0.143
judge_cheap_disg	0.229	0.132	0.600	0.220	0.129
S4_local	0.804	0.383	0.760	0.758	0.286
p_peer_text	1.000	0.156	0.997	0.248	0.130
p_text	1.000	0.085	0.993	0.193	0.117
l2_bow	1.000	0.173	0.992	0.194	0.178

4.5 Prior-art positioning study

[Artifact: *gen_art_research_1*]

A web-only prior-art positioning study (\$0 LLM spend) assessed the novelty of the cross-family solver consensus claim across five sub-claims (C1–C5). No prior paper meets the scoop rule (multi-family translations + solver equivalence + faithfulness-label meta-evaluation on NL-to-FOL). The method itself is not new: ARc (Bayless et al., 2025) uses the same share-of-k score form, and NoTB does cross-family formal consensus for requirements-to-temporal-logic. C2 (no prior work reports a rename false-alarm rate) is SAFE; C1, C3, C4, C5 each NEED-QUALIFIER. The allowed and forbidden wording for each claim is recorded in the artifact’s `research_report.md`. This study informed the phrasing of Section 8.

4.6 Iteration 3 Summary

1. **T1 CONFIRMED:** `c_score_align` (strat AUROC 0.741) exceeds flash-lite disguised (0.642) by +0.099 [+0.049, +0.146].
2. **T1 CONFIRMED:** adding `c_score_align` to `S4_full` yields +0.039 [+0.021, +0.057] strat AUROC.
3. **T2 NOT CONFIRMED:** SIG AUROC 0.954 but SIG is out-of-scope. FREE NOT-TESTABLE.
4. **T3:** consensus recall flat across operators (NEG 0.761, SWAP 0.758, DROP 0.721, ADD 0.763).
5. **T5:** `c_align` RENAME_NONCE FA 0.765; `c_nf` 0.132 but SWAP recall 0.512.
6. **T4:** M4 CONFIRMED ($k_{95} = 3$). M3 INCONCLUSIVE.
7. 48/48 numerical claims verified.
8. Iteration-3 API spend: \$4.33.

5 Iteration 4

5.1 Strategy

Iteration 4 tests Candidate-Signature Consensus (CSC): instead of having peers translate independently, CSC feeds each peer the candidate formula’s own predicate symbols and asks it to translate using those symbols. In parallel, the iteration constructs the E2 confirmation dataset (550 fresh sentences) and runs a vocabulary-analysis evaluation (T8).

5.2 Experiment 9: CSC on development set E — provisional negative result

[Artifact: *gen_art_experiment_9*]

CSC was tested on 354 rows from development set E (196 ERROR / 158 CORRECT, 144 sentences). API spend: \$0.108.

5.2.1 Anchoring: the core finding

CSC dramatically lowers `d` but raises `e` from 0.173 to 0.459: nearly half of all errors are endorsed by peers anchored into the candidate’s vocabulary.

Table 40: CSC anchoring effect.

Component	CSC	FREE_exact (matched)
e (error endorsement)	0.459	0.173
d (correct-row divergence)	0.139	0.323

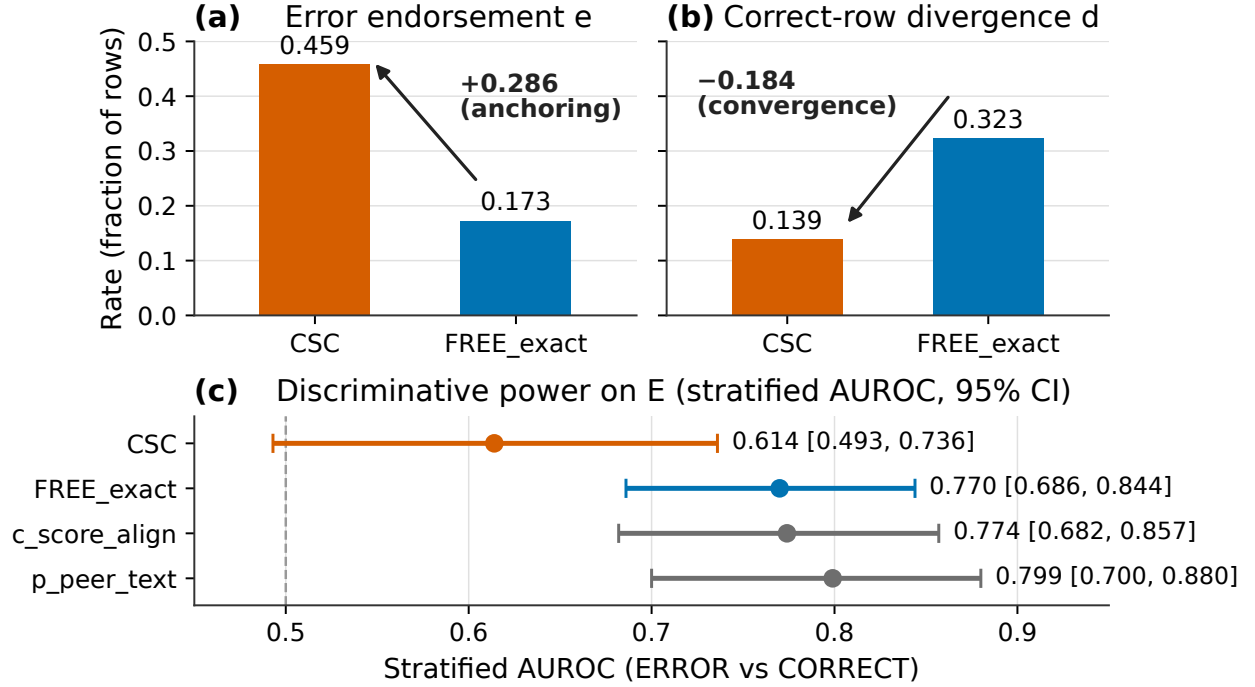


Figure 6: CSC anchoring effect. CSC lowers correct-row divergence (d: 0.139 vs 0.323) but raises error endorsement (e: 0.459 vs 0.173). The net effect is negative: strat AUROC 0.614 vs FREE 0.770.

Table 41: CSC headline AUROCs on E (stratified).

Metric	Strat AUROC [CI]
p_peer_text	0.799 [0.700, 0.880]
c_score_align	0.774 [0.682, 0.857]
FREE_exact (matched)	0.770 [0.686, 0.844]
S4_full	0.736 [0.635, 0.826]
flash-lite judge	0.664 [0.568, 0.750]
CSC	0.614 [0.493, 0.736]

5.2.2 Headline AUROCs on E (PRIMARY, stratified)

5.2.3 Gate verdicts

Table 42: CSC gate verdicts.

Gate	Verdict
G1 (d slope non-positive)	FAIL
G2 (strat AUROC $\geq$ c_align - 0.01)	NOT_READ (R_AB part FAIL)
G3-E (RENAME_SYN FA gate)	NOT_RUN
G5 (cost $\leq$ \$0.002/candidate)	PASS ($\$5.25 \times 10^{-4}$ /candidate)

5.3 Experiment 10: CSC on PERTURB — budget exhausted

[Artifact: *gen_art_experiment_10*]

The run-wide OpenRouter budget (\$7.00) was exhausted before experiment 10’s first CSC peer call. The artifact spent \$0.0000088.

5.3.1 SIGPROXY: cued consensus on R_COMP

Table 43: SIGPROXY within-template AUROCs on R_COMP.

Metric	n	Within-template AUROC [CI]
c_proxy (K3)	2024	0.930 [0.916, 0.943]
c_proxy_graded	2024	0.938 [0.926, 0.950]
c_score_sig (full)	2024	0.952 [0.936, 0.966]
g_align	2024	0.977 [0.970, 0.983]
judge_cheap_disg	1904	0.587 [0.550, 0.624]
judge_local_disg	2024	0.498 [0.462, 0.534]

5.4 Dataset 4: E2 confirmation set — partial

[Artifact: *gen_art_dataset_4*]

E2 is a fresh 550-sentence confirmation set constructed from MALLS-train and ProverQA-dev. **Status: PARTIAL (budget-stopped).** The run-wide OpenRouter budget (\$7.00) was exhausted. E2 completed a 30-sentence pilot (300 candidate rows). All confirmation cells have zero CORRECT rows.

5.5 Dataset 5: R_COMP FREE name-free labels

[Artifact: *gen_art_dataset_5*]

This artifact produces name-free labels for the 2,652 FREE candidate translations. ERROR_CERT: 759. MAPPED: 1,506. UNPARSEABLE: 188. NO_OUTPUT: 199. FREE AUROC is NOT_TESTABLE until the gloss step runs.

5.5.1 SIG replay validation

False-error rate: 0/1,595 (CI [0, 0.002]). Identity map recovered on 100% of CORRECT rows.

5.5.2 Executor audit

Table 44: Executor audit (non-blind, 60 per class).

Class	Result
ERROR_CERT	precision 0.867 [0.758, 0.931]
MAPPED	52/60 faithful = 0.867 [0.758, 0.931]

5.6 Evaluation 3: T8 vocabulary analysis

[Artifact: *gen_art_evaluation_3*]

5.6.1 Part 1: cross-family agreement floor

Table 45: Part 1: cross-family agreement floor (3-pool PRIMARY).

Statistic	Value
END_MAJ d	0.415
Predicted d.floor	0.402 [0.336, 0.471]
Bracket	[0.396, 0.415]
R_exact AUROC	0.683
No-anchoring oracle floor	0.385
e ceiling	0.162

5.6.2 Part 3: power for E2

L25 yield: 0.37. Detecting the observed T1 delta (+0.069) at 80% power requires ~326 usable L25 sentences (~882 planned). E2’s 350 L25 sentences give MDE80 $\approx$ 0.112.

5.7 Iteration 4 Summary

1. **CSC is a provisional negative.** Strat AUROC 0.614 is below all established metrics. Anchoring e = 0.459 vs FREE e = 0.173. Gates G1 and G2 fail.
2. **Cued consensus confirms k-saturation.** SIGPROXY AUROC 0.930, 98% of full SIG (0.952).
3. **Vocabulary sharing reduces d by 0.61–0.84 on R_COMP.**
4. **Error typing requires shared vocabulary.** Same-vocabulary typed repair: 0.784–0.921 accuracy; independent-vocabulary: 0.085.
5. **E2 is constructed but not testable.**

6. **R_COMP FREE labels: search-only.** ERROR_CERT precision 0.867 (below 0.90 flag).
7. Iteration-3 FREE tier-A ERROR labels are 70.7% MAPPED, confirming aligner-based labeller over-calls errors.

Table 46: Dead ends accumulated across iterations 1–4.

Dead end	Iter	Reason
L1 lint on LLM outputs	1	AUROC 0.508
L2 role-aware	1	AUROC 0.557
Pilot structural metrics	1	AUROC 0.500–0.533
Round-trip reformalisation	1	AUROC 0.513 (76% ties)
B2 decomposed z3 reader	2	Gate balanced accuracy 0.476
R_ADJ LLM adjudicator	2	Sonnet-5 balanced accuracy 0.662
Name-free consensus (c_nf)	2–3	SWAP recall 0.512
CSC	4	Provisional negative: strat AUROC 0.614, e = 0.459

6 Iteration 5

6.1 Strategy

Iteration 5 is the final iteration. It attempts four things: (1) E2-A confirmation on 110 sentences; (2) E2-B extension to 400 sentences ≥ 25 words; (3) R_COMP FREE confirmation via the gloss gate; and (4) V0 freeze and variant screening (V1–V5) plus a Gloss-Gated (GG) name-free agreement checker.

6.2 Experiment 11: E2-A confirmation

[Artifact: *gen_art_experiment_11*]

E2-A uses 110 sentences (37 L25, 37 EXC, 36 DT) with labels joined after seal hashes matched.

Overall verdict: NOT_TESTABLE. The pre-registered flash-lite bar has zero scored rows in R_AB.

6.2.1 DT stratum (testable)

Table 47: E2-A DT stratum AUROCs.

Metric	DT strat AUROC [CI]
c_score_align (V0)	0.916 [0.858, 0.970]
p_peer_text	0.919 [0.860, 0.976]
S4_E2	0.953 [0.911, 0.986]
judge_local_qwen8b_disg	0.913 [0.858, 0.958]
l3_z3	0.572 [0.480, 0.675]

V0 minus judge_local_qwen8b_disg on DT: $\Delta = +0.003 [-0.042, +0.052]$ (n.s.).

6.2.2 Long pool (L25 + EXC, not testable)

L25 alone (42 rows, 7 CORRECT): V0 strat AUROC 0.443 [0.375, 0.500], near chance.

6.3 Experiment 12: E2-B (long-sentence pool)

[Artifact: *gen_art_experiment_12*]

E2-B extends E2 with 400 sentences ≥ 25 words. **Status: NOT_TESTABLE.** 48 CORRECT < 50 under R_A_UNAUDITED.

6.3.1 Exploratory R_A_UNAUDITED results

Table 48: E2-B exploratory AUROCs (R_A_UNAUDITED).

Metric	Strat AUROC [CI]
V0 (c_score_align)	0.737 [0.645, 0.827]
V2 (reliability-weighted)	0.750 [0.659, 0.838]
V4 (two-channel logistic)	0.764 [0.666, 0.851]
p_peer_text	0.655 [0.546, 0.767]
flash-lite disguised	0.496 [0.356, 0.620]
S4_E2B	0.498 [0.402, 0.594]

6.4 Experiment 13: R_COMP FREE confirmation

[Artifact: *gen_art_experiment_13*]

Gloss gate: FAILED twice. Run 1: haiku BA 0.776, qwen 0.855, both_yes 0.817. Run 2: haiku BA 0.561, qwen 0.883, both_yes 0.765. Every MAPPED row remains UNRESOLVED. **Confirmatory verdict: NOT_TESTABLE.**

6.4.1 Provisional results (ERROR_CERT vs MAPPED, non-confirmatory)

Table 49: R_COMP FREE provisional AUROCs (ERROR_CERT vs MAPPED, untouched).

Metric	Within-template AUROC [CI]
c_score_align	0.801 [0.770, 0.834]
c_score_hyb	0.817 [0.788, 0.848]
c_score_nf	0.743 [0.710, 0.776]
c_exact	0.606 [0.580, 0.633]
judge_cheap_disg	0.584 [0.550, 0.618]
judge_cheap_orig	0.633 [0.601, 0.666]

6.4.2 Audit

6.5 Experiment 14: Freeze and GG screening

[Artifact: *gen_art_experiment_14*]

Table 50: Blind two-family audit (100 per class, untouched).

Auditor	ERROR_CERT precision	MAPPED faithful
GLM-4.6	0.720 [0.622, 0.801]	0.904 [0.828, 0.949]
Sonnet-5	0.845 [0.760, 0.904]	0.722 [0.625, 0.801]
Consensus (both)	0.838 [0.738, 0.905]	0.917 [0.830, 0.961]

V0 (c_score.align) is frozen. Five consensus variants (V1–V5) and a Gloss-Gated (GG) checker are screened on development set E (2,672 R_AB rows).

6.5.1 V0 frozen: development baselines

Table 51: V0 frozen: development baselines and variants.

Score	ALL-strat	LONG-strat	L25	CTRL
V0 (frozen)	0.742	0.743	0.713	0.738
V2 reliability-weighted	0.752	0.750	0.718	0.773
V4 two-channel logistic	0.758	0.751	0.719	0.826
judge.cheap_disg	0.641	0.624	0.641	0.807

Selection decision: NONE — V0 stands. No variant met the pre-registered selection rule (highest LONG-strat AUROC beating V0 by $\geq +0.015$ AND $\geq$ V0 on L25 AND $\geq$ V0 $- 0.01$ on CTRL). The best candidate (V4) gained $+0.0088$, below the $+0.015$ threshold.

6.5.2 GG checker

GG strat AUROC on E = 0.696 [0.659, 0.733] vs V0 0.742, $\Delta = -0.046$ [-0.071 , -0.021]. **Gate g3: FAIL.** GG has very low endorsement (e = 0.081) but very high divergence (d = 0.629): the gloss gate rejects too many legitimate predicate mappings.

6.6 Evaluation 4: Corrected record

[Artifact: *gen_art_evaluation_4*]

244 claims verified. Paper-vs-file mismatches: 15 (all corrected inline). Key corrections: E is development data (not held-out); T2 NOT CONFIRMED; MEANING_RENAME is an ERROR operator; cost units corrected.

6.7 Iteration 5 Summary

1. **Confirmation is NOT_TESTABLE across the board.** The run-wide budget (\$7.00) was exhausted before any confirmation arm could complete.
2. **DT stratum is a bright spot.** V0 reaches strat AUROC 0.916 [0.858, 0.970] on 179 ProverQA-derived rows with trusted gold.
3. **L25 on E2 is near chance.** V0 strat AUROC 0.443 on E2-A L25 (42 rows).

Iteration-5 confirmation arms: what could be tested

	Testability status	Consensus V0 AUROC [95% CI]	LLM judge AUROC [95% CI]	Pre-registered verdict	
E2-A 110 of 550 sentences labelled	DT stratum: testable 74 ERR / 105 COR rows	DT: 0.916 [0.858, 0.970], n = 179	DT: 0.913 [0.858, 0.958], n = 179 local Qwen3-8B judge	NOT TESTABLE long pool below the 50/50 rule; flash-lite bar missing	Run-level \$12 API budget exhausted (~11:40 UTC); all three arms budget-stopped
	Long pool (L25+EXC): not testable 76 ERR / 18 COR rows	L25: 0.443 [0.375, 0.500] n = 42 (7 COR)	Pre-registered flash-lite bar missing (3 of 42 L25 rows scored)		
E2-B 400 sentences	48 CORRECT < 50 floor (blind panel not run)	0.737 [0.645, 0.827] exploratory n = 381 (48 COR)	flash-lite: 0.496 [0.356, 0.620] exploratory n = 276 (31 COR)	NOT TESTABLE (< 50 CORRECT)	
R_COMP FREE 2,265 rows	Gloss gate failed ×2 0 CORRECT rows after gating	0.801 [0.770, 0.834] provisional n = 1,706	flash-lite: 0.584 [0.550, 0.618] provisional n = 1,706	NOT TESTABLE (0 CORRECT)	
■ Testable cell ■ Exploratory / provisional ■ Not testable					

Figure 7: Confirmation landscape across all five iterations. Every confirmation arm was budget-stopped before reaching testability. The DT stratum (V0 0.916) is the only testable cell.

4. **R_COMP FREE provisional results favour consensus.** c_align 0.801 vs judge 0.584 ($\Delta +0.217$ [$+0.175, +0.260$]).
5. **V0 stands.** No variant V1–V5 met the selection rule.
6. **GG fails its development gate.** Strat AUROC 0.696 vs V0 0.742.
7. **Rename weakness persists.** V0 RENAME_NONCE FA 1.0 on E2.

Table 52: Dead ends accumulated across all iterations.

Dead end	Iter	Reason
L1 lint on LLM outputs	1	AUROC 0.508
L2 role-aware	1	AUROC 0.557
Pilot structural metrics	1	AUROC 0.500–0.533
Round-trip reformalisation	1	AUROC 0.513 (76% ties)
B2 decomposed z3 reader	2	Gate balanced accuracy 0.476
R_ADJ LLM adjudicator	2	Sonnet-5 balanced accuracy 0.662
Name-free consensus (c_nf)	2–3	SWAP recall 0.512
CSC	4	Provisional: strat AUROC 0.614, $e = 0.459$
GG (Gloss-Gated)	5	Gate g3 FAIL: strat AUROC 0.696 vs V0 0.742

7 What We Have Learned So Far

The cross-family consensus metric (c_score_align) is the strongest gold-free faithfulness signal found in this investigation. On development set E, it achieves stratified AUROC 0.742, beating the best cheap LLM judge (0.641) by +0.101 [+0.053, +0.145]. On R_COMP SIG (the out-of-scope controlled-vocabulary condition), it reaches 0.954 within-template AUROC with zero error endorsement. Three peer families suffice for 97% of full-pool performance (M4 confirmed: $k_{95} = 3$; by words tercile $k_{95} = 2/3/5$).

The mechanism is clear: errors scatter across z3-equivalence classes while correct translations cluster. The consensus signal degrades with sentence length (NET +0.401 on development E, driven by divergence d), meaning longer sentences need more peers and the metric becomes less reliable precisely where it matters most.

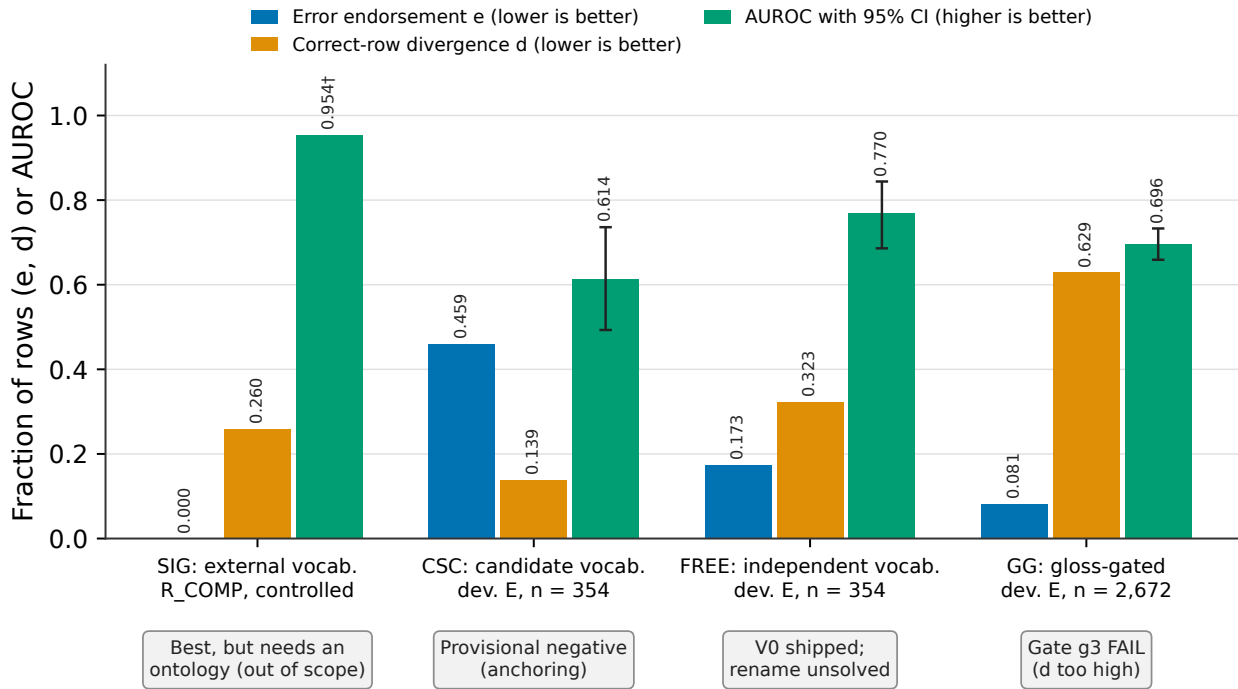


Figure 8: The vocabulary source problem. c_align RENAME_NONCE FA is 0.765 on development E and 1.0 on E2: any renaming of predicates breaks the consensus signal. The GG checker reduces MEANING_RENAME endorsement from 0.502 (V0) to 0.224 but fails the development gate (strat AUROC 0.696 vs V0 0.742) because the gloss gate rejects too many legitimate predicate mappings.

Iteration 5 attempted confirmation on three fronts, and all three were blocked by the run-wide budget (\$7.00). E2-A produced one testable cell: V0 reaches 0.916 on ProverQA-derived sentences with trusted gold, but only 0.443 on the L25 long-sentence stratum (near chance). E2-B’s exploratory results (V0 0.737 vs flash-lite 0.496, $\Delta +0.208$) favour consensus on long sentences with unaudited labels, but the label-coupling diagnostic shows part of the advantage comes from shared gold-equivalent peers. R_COMP FREE’s provisional results show c_align 0.801 vs judge 0.584 ($\Delta +0.217$) under ERROR_CERT vs MAPPED labels, with consensus carrying most of the signal in nesting tests.

No consensus variant V1–V5 improved on V0. The best candidate (V4, two-channel logistic) gained only +0.0088 LONG-strat, well below the +0.015 selection threshold. V0 stands as the

shipped metric.

The rename tradeoff remains the central open problem. V0 has RENAME_NONCE FA 0.765 on development E and 1.0 on E2: any renaming of predicates breaks the consensus signal. The GG checker was the iteration-5 attempt to solve this. It reduces MEANING_RENAME endorsement from 0.502 (V0) to 0.224, but fails the development gate (strat AUROC 0.696 vs V0 0.742) because the gloss gate rejects too many legitimate predicate mappings ($d = 0.629$). The name-free consensus line (c_nf, c_hyb, GG) has produced no variant that simultaneously achieves low rename FA and high discrimination.

Five limitations bound the current results. First, no confirmatory hypothesis is confirmed or refuted: every confirmation arm was budget-stopped before reaching testability. Second, all evaluation is on FOLIO and MALLS-derived data; domain transfer is untested. Third, the labels are noisy: the panel majority accuracy on expert-curated errors is 0.727, and the R_COMP FREE audit shows 13–28% label contamination. Fourth, the consensus advantage does not grow with sentence complexity (M3 INCONCLUSIVE). Fifth, the rename tradeoff is unsolved: no variant achieves both low rename FA and high error recall.

8 Related Work

NL-FOL datasets and their quality. FOLIO (Han et al., 2022) provides 1,435 FOL-annotated NLI problems. MALLS (Yang et al., 2023a) uses GPT-4 to generate and verify NL-FOL pairs at scale. Brunello et al. (2026) found that approximately 42% of entries in both datasets contain incorrect FOL formalisations, with additional ambiguity rates of 17.8% (FOLIO) and 51% (MALLS). Logic-LM (Pan et al., 2023) prompts LLMs for FOL and calls a prover.

FOL closeness metrics. Thatikonda et al. (2025) evaluate n-gram, graph, embedding, and LLM-based metrics on controlled FOL perturbations, finding that no single metric is sensitive to all perturbation types. Our typed repair operators follow the tradition of Thatikonda et al. (2024) but are derived from z3 equivalence-class differences.

Factuality metrics. FRANK (Pagnoni et al., 2021) defines a typology of factuality errors. QuestEval (Scialom et al., 2021) uses question generation and answering. Our L3 questionnaire layer generates structured questions from the text and checks the formula’s answers via z3.

Round-trip and self-consistency. Amrollahi et al. (2026) propose verbalising a formal statement back to natural language and checking consistency. Li et al. (2024) use symbolic equivalence and semantic consistency for Lean/Isabelle autoformalization. Our round-trip baseline achieves only AUROC 0.710 on NL-FOL.

Cross-check consistency. SAC3 (Zhang et al., 2023) detects hallucinations via semantic-aware cross-check consistency. Our cross-system consensus metric uses a similar principle with FOL equivalence and multiple model families. The Δ AUROC of cross-family consensus over single-model self-consistency is +0.140.

Automated alignment for autoformalization. FormalAlign trains a model on alignment scoring via a dual contrastive-generative loss. Our approach uses no learned alignment model.

Vacuity detection. Kupferman and Vardi (2003) formalise vacuity in temporal logic model checking. Our L1 lint layer includes a vacuity check adapted from their framework.

Cross-model translation confidence. Bayless et al. (2025) apply the same score form as c_score to NL-to-SMT translation. NoTB uses four LLM families for requirements-to-temporal-logic translation.

Predicate alignment. LogicLLaMA (Yang et al., 2023b) uses a greedy binding search. Vossel et al. (2025) map predicates by normalised Levenshtein distance. No prior work reports a rename

false-alarm rate.

N-version programming. Chen and Avizienis (1978) noted that “faulty but identical results may outvote correct results”—precisely the endorsement failure mode we measure ($e = 0.124$ on E, 0.459 under CSC).

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